



School of Future Tech

Case Study Report

on

106.Math Quiz Battle Arena (Competitive Quiz Platform)

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1. Introduction to the Case Study

Mathematics practice platforms are often limited to static question banks without interactive competition or progress tracking. Students require engaging systems that allow them to test their knowledge, monitor performance, and compete with others.

This case study focuses on designing and implementing a **Math Quiz Battle Arena**, a web-based quiz platform developed using **HTML, CSS, and JavaScript**. The application allows users to attempt timed math quizzes across different difficulty levels, track their scores, view rankings on a leaderboard, and interact with other users through a discussion forum.

The project demonstrates how modern web technologies can be used to create an interactive educational application that runs entirely on the frontend.

2. Problem Statement / Case Background (Abstract)

Background

Many online quiz systems lack real-time feedback, difficulty levels, and competitive ranking systems. Users are unable to track their performance effectively or compare results with other players.

Abstract

This case study presents the design and development of a **Math Quiz Battle Arena web application** that provides:

- Timed math quizzes
- Difficulty levels (Easy, Medium, Hard)
- Real-time scoring system
- Leaderboard ranking system
- Interactive forum for discussion

The system generates random math questions and validates user answers instantly. Scores and completion time are stored using **LocalStorage**, allowing the leaderboard to display top players. The application runs entirely in the browser without requiring a backend server.

3. Case Study Design

The Math Quiz Battle Arena is designed as a **multi-page web application** consisting of several modules.

⇒ Application Pages

The system contains the following pages:

- Home Page – Introduction to the quiz platform
- Levels Page – Select difficulty level and enter player name
- Quiz Page – Displays math questions and answer options
- Result Page – Shows final score and performance message
- Leaderboard Page – Displays top players and rankings
- Forum Page – Allows users to post and read comments

⇒ Navigation System

A navigation bar allows users to move between pages easily:

Home → Play → Leaderboard → Forum

⇒ Data Storage

The system stores important data using **LocalStorage**:

- Player name
- Quiz score
- Time taken
- Leaderboard records
- Forum comments

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

⇒ Programming Technologies Used

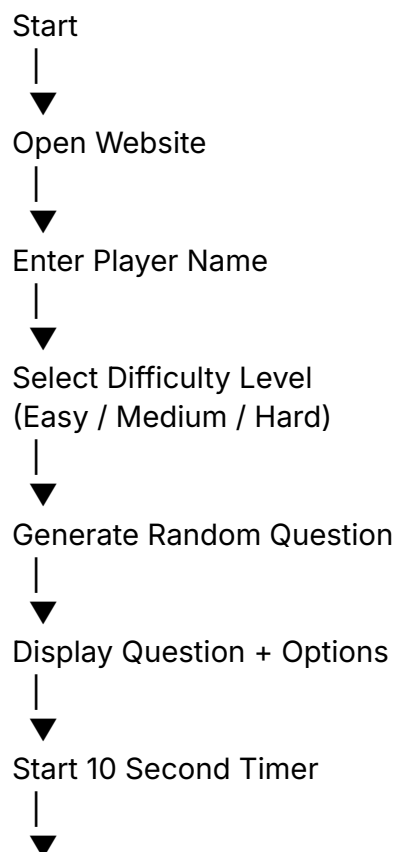
- **HTML5** – Structure of the web pages
- **CSS3** – User interface design and layout
- **JavaScript** – Application logic and interactivity
- **LocalStorage API** – Data storage in browser
- **Google Fonts** – UI typography

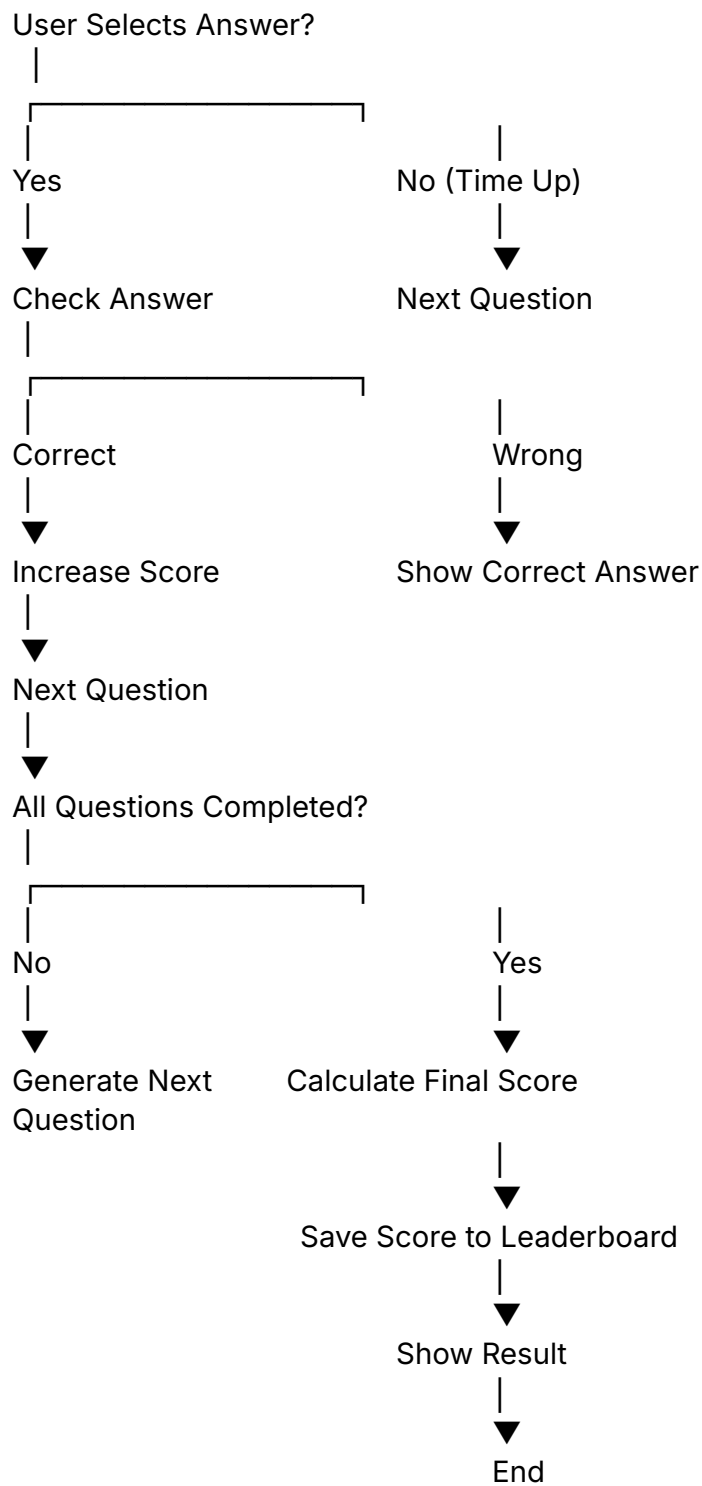
⇒ Concepts Applied

The application uses the following programming concepts:

- Random number generation
- Conditional statements (if-else)
- Loops for quiz progression
- Event handling for button clicks
- Timer functions using setInterval()
- LocalStorage for persistent data storage

⇒ Algorithm





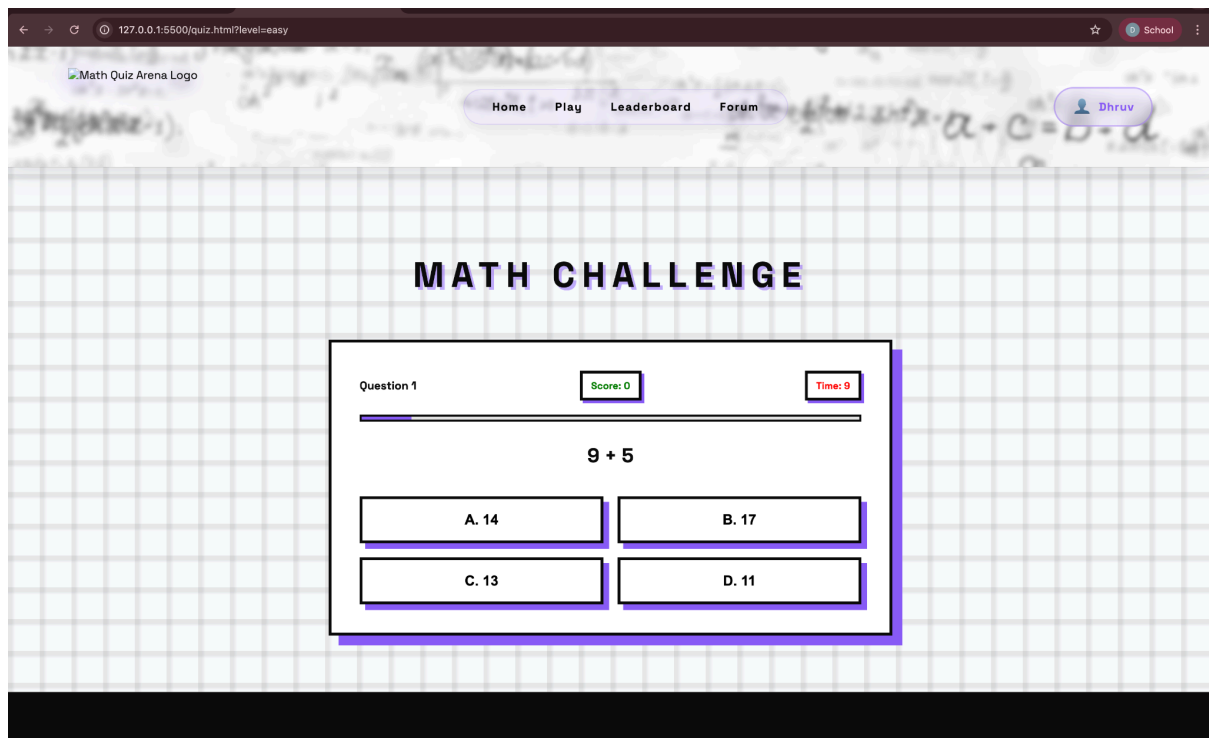
5. Problem Statement / Case Study Implementation Details and Snapshots.

⇒ Quiz Module

- The quiz module generates math questions dynamically depending on the selected difficulty level.
- Each question has four options, and users must answer within **10 seconds**.

⇒ Features

- Random question generation
- Difficulty level system
- Countdown timer
- Real-time score update



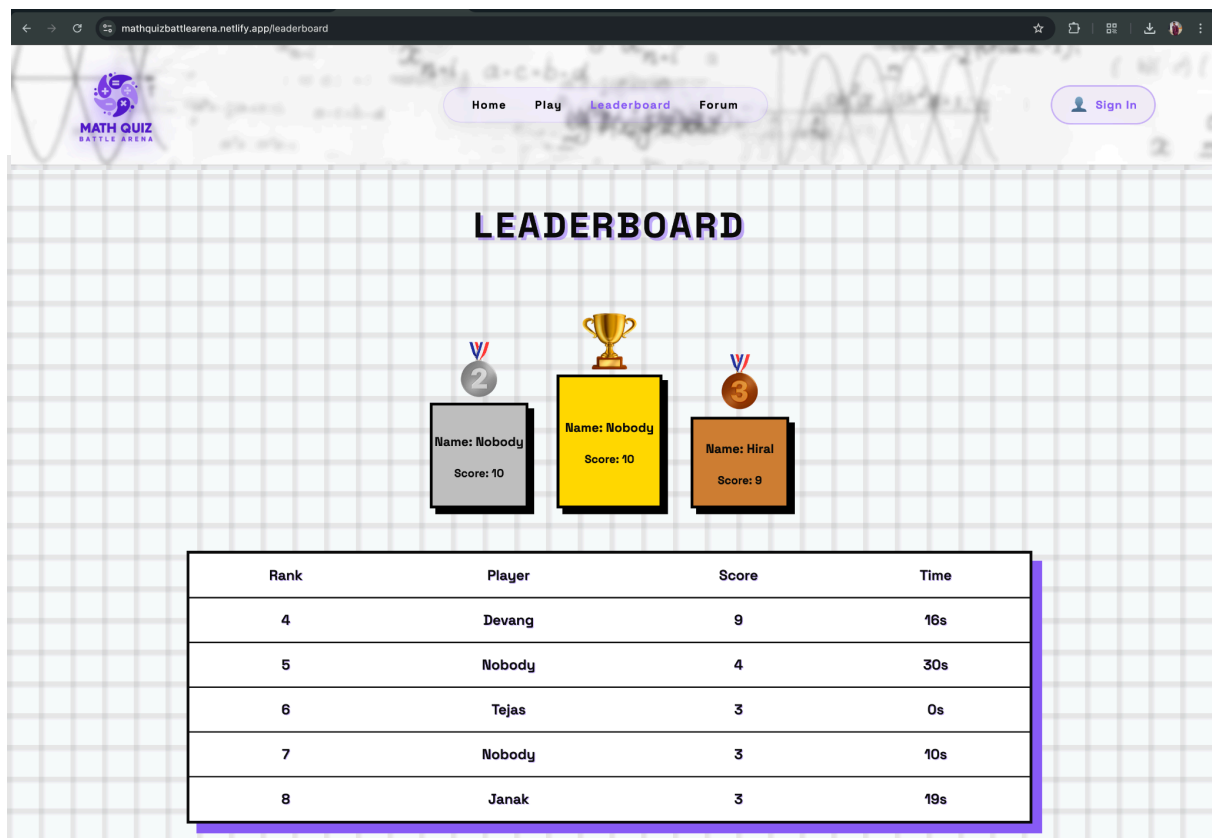
=> Leaderboard Module

The leaderboard ranks players based on:

=> Highest score

=> Lowest completion time

Player data is stored in **LocalStorage** and displayed in a table format.



The screenshot displays the 'LEADERBOARD' page of the 'MATH QUIZ BATTLE ARENA' application. The page features a navigation bar with 'Home', 'Play', 'Leaderboard', and 'Forum' links, and a 'Sign In' button. The main content area shows a podium with three positions:

- Rank 2: Name: Nobody, Score: 10
- Rank 1: Name: Nobody, Score: 10
- Rank 3: Name: Hiral, Score: 9

Below the podium, a table lists the following players:

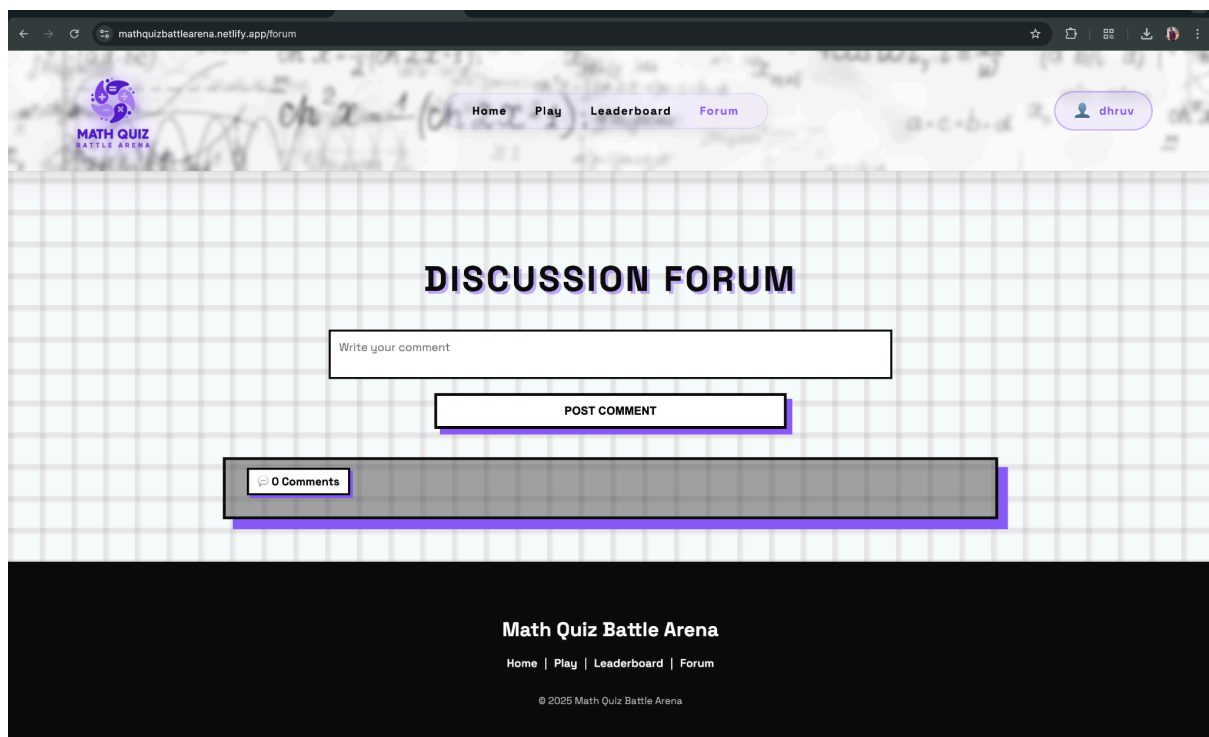
Rank	Player	Score	Time
4	Devang	9	16s
5	Nobody	4	30s
6	Tejas	3	0s
7	Nobody	3	10s
8	Janak	3	19s

=> Forum Module

The forum allows users to share messages and discuss math strategies.

=> Features

- Comment posting system
- Comment Counter
- LocalStorage data persistence



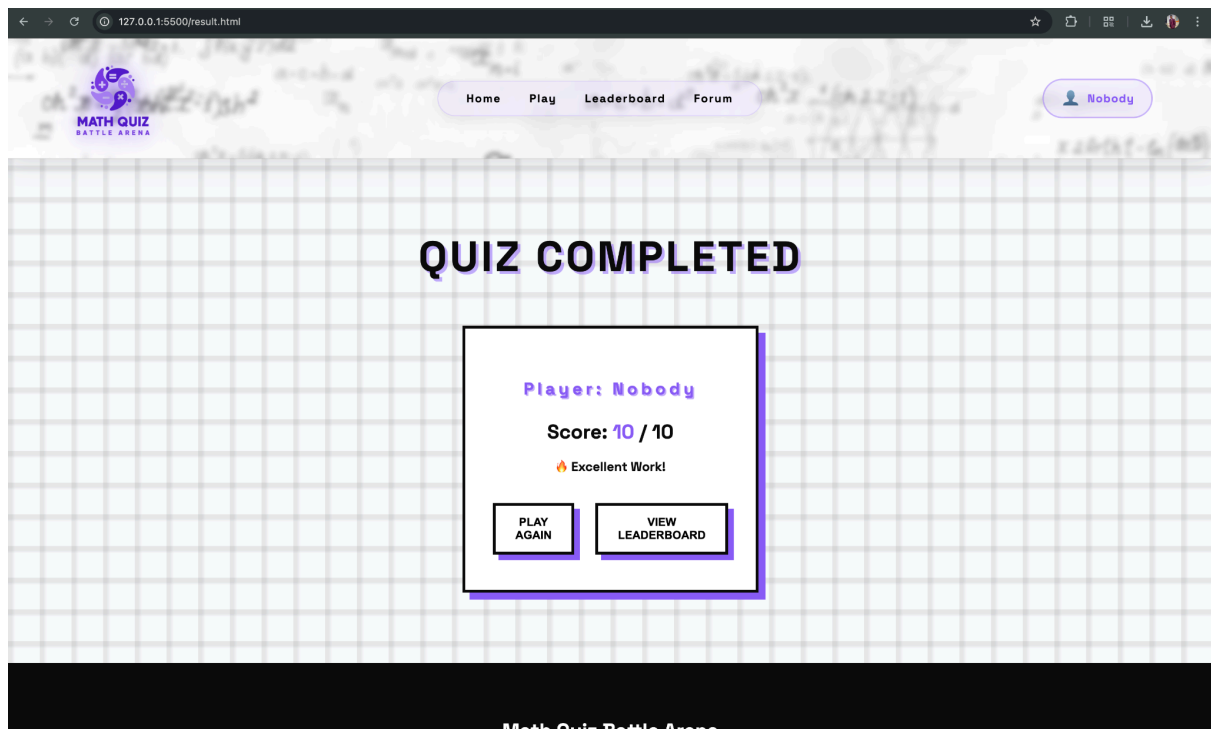
=> Result Module

After completing the quiz, the result page shows:

- Final score
- Player name
- Performance message

Example messages:

- 🔥 Excellent Work
- 👍 Good Job
- 💪 Keep Practicing



6. Problem Statement / Case Study Results and Conclusion.

⇒ Results

The application successfully demonstrates the implementation of an interactive quiz platform with the following outcomes:

- Random math question generation works correctly
- Real-time scoring system functions accurately
- Leaderboard ranking updates dynamically
- Forum allows user interaction
- LocalStorage maintains user data across pages

⇒ Conclusion

This case study successfully implements a **Math Quiz Battle Arena web application** using HTML, CSS, and JavaScript. The system provides an engaging environment where users can test their math skills, compete on leaderboards, and discuss solutions with others.

The project demonstrates how frontend technologies can be combined to create a complete educational application without requiring backend infrastructure.

7. References

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